Text Mining Term Project

Sentiment Analysis on Drug Reviews

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Introduction

- Sentiment Analysis on Drug Reviews
- Data Set: drugs.com
- Text Mining Methods + Machine Learning & Deep Learning Models
- Predicting Whether Reviews are Positive,Negative or Mixed Sentiment

Motivation and Goal

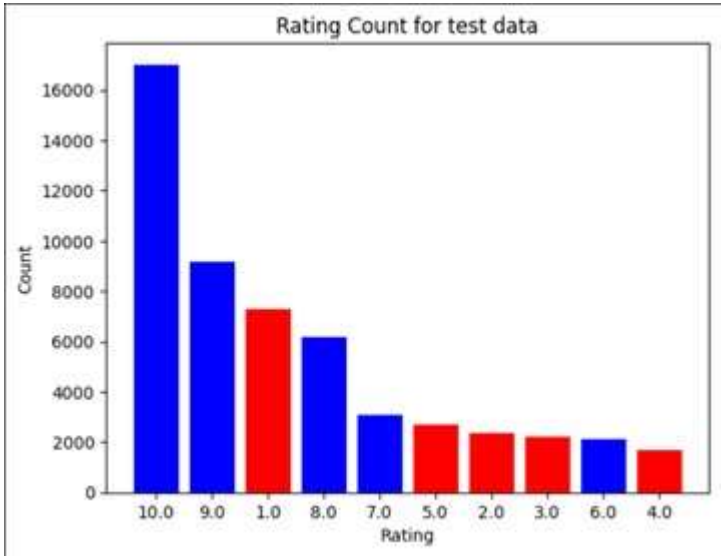
- Motivation:
  - Sentiment Analysis: one of the most important NLP Task
- Challenge:
  - Capturing the semantic of a document
  - Developing Robust Models to obtain high evaluation metrics

Dataset Overview

Train Set : (161297, 7), Test Set: (53766, 7)

	Unnamed: 0	drugName	condition	review	rating	date	usefulCount
0	206461	Valsartan	Left Ventricular Dysfunction	"It has no side effect, I take it in combinati...	9.0	May 20, 2012	27
1	95260	Guanfacine	ADHD	"My son is halfway through his fourth week of ...	8.0	April 27, 2010	192
2	92703	Lybrel	Birth Control	"I used to take another oral contraceptive, wh...	5.0	December 14, 2009	17

Dataset Overview



Methodology

Preprocessing:

- Conversion of HTML elements
- Lowercase conversion
- Tokenization
- Removal of Non-Alphanumerics, frequent and rare words, and stopwords
- Class Mapping

Feature Extraction:

- Used TF-IDF for Traditional Machine Learning models.
- Used Word2Vec for advanced Deep Learning algorithms.
- And used Transformer-based models such as BERT.

Methodology

Models:

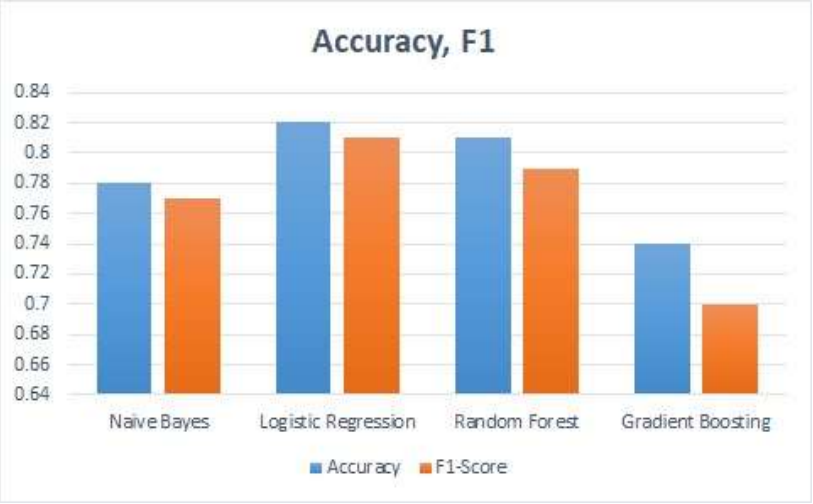
- Traditional ML Models:
- Naive Bayes, Logistic Regression, Random Forest, Gradient Boosting
- Deep Learning Models:
- CNN, LSTM, BERT

Interpretations:

- Comparing Models' Performance based on metrics
- Extracting false predictions and understanding the reasons behind

Results and Evaluation

- Feature Extraction Method: TF-IDF
  - Up to 5000 features, including bi-gram windows
  - Models: (Binary Classification)
    - Logistic Regression
      - Precision: .81, Recall: .82, F1: .81
    - Naive Bayes
      - Precision .78, Recall: .78, F1: .77
    - Random Forest
      - Precision: .81, Recall: .81, F1: .79
    - Gradient Boosting
      - Precision: .75, Recall: .74, F1: .70



Results and Evaluation

- Feature Extraction Method: Word2Vec
  - Static Word Embeddings
  - Vector Size: 50, 100, 200 (100 optimum)
  - Assumed to capture semantic meaning better
  - Deep Learning Models (Binary Classification):
    - LSTM: Precision: .83, Recall: .82, F1: 83
    - CNN: Precision: Precision: .78, Recall: .78, F1: .78

Results and Evaluation

- BERT:
  - Bidirectional Encoder Representations Transformers
  - Contextualized Embeddings
  - Transformer Based Model
  - Pretrained BERT on lowercase English Text
  - Binary Classification
  - Three Class Classification + Class Balancing

Results and Evaluation

- BERT & Binary Classification:
  - Precision: .92, Recall: .92, F1: .92

- BERT & Three Class Classification
  - Class Balancing with random under sampling
  - Negative, Mixed Sentiment (or Neutral), Positive
    - Precision: .82, Recall: .81, F1: .81

Discussion

- More naive representation + simple models
  - Provides promising results
  - Outperforms Deep Learning Models + Word2Vec
    - Possible Justifications:
      - Small Vocabulary Diversity: ~44k in ~165k reviews
      - Average token size of a review: ~40 tokens
      - ***Indicating narrow semantic context and simple narrative***

Discussion

- BERT:
  - Outperforms all other methods & models in binary classification
    - Powered by transformer technology
    - Better at capturing contextual meanings
    - Already pre-trained on large corpus
  - Even capable of capturing mixed sentiment (neutral) reviews
    - “Not Bad” → Predicted as Neutral whereas it was predicted as good in binary classification
    - .82 F1 Score

Discussion

Even BERT can miss context in some cases:

"I was on this for 5 years (and birth control pills for about 12 years), and would have told you how fabulous it was. <List all the benefits everyone else has listed, here.> Then a friend of mine convinced me to stop birth control all together for a while, and I turned into a new and better person! I dropped 5 lbs instantly, had a huge sex drive, more lubrication, and just felt better. I wouldn't say I was depressed before, but after going off of it, I would describe my time on birth control as feeling like I was inside a cloud. Less emotions in general, sometimes a little down. I'm bummed I'm back on it, but it seems to be the least impactful than the others. I might try Paragard to avoid the hormones.",

True Label: Neutral, Predicted Label: Positive

"I am a 12 year survivor of mirapex/Pramepexole. It was a miracle to begin with. I only slept about 4hrs a night but had enormous energy and never tired. That lasted nine years. But during that time I lost bladder control, impulse disorders, loss of concentration, myalgia neuropathy, falling asleep while driving, many more and augmentation. I was on .5 mg. Detoxing med free from this med was horrendous! After detox most of the side effects cleared up. But I still have issues with memory and concentration. This is the only med I took on a reg bases as my health was good. Im still suffering the side effects of the Withdrawl from it. Studies should be for longer than 12 weeks."

True Label: Negative, Predicted Label: Positive

Conclusion

- Sentiment Analysis
- Drug Reviews & Ratings (drugs.com)
- Utilized different representation methods
- Different Machine Learning and Deep Learning Models
- BERT: Winner but with some weaknesses

THANKS FOR YOUR ATTENTION